$1+2 = n$

$TC = O(2^n)$

```
int towerOfHanoi(int n, string A, string C, string B) {
    if (n == 0) {
        return 0;
    }

    int result = towerOfHanoi(n - 1, A, B, C);
    cout << "Move the " << n << "-th disk from " << A << " to " << C << endl;
    result++; // To move nth disk from A to C
    result += towerOfHanoi(n - 1, B, C, A);

    return result;
}
```

(n = 3, A, C, B)

(2, A, B, C)

+1

1 1

(1, A, C, B) (1, - )

+1

0 0

(0, A, B, C) (0 ... )

| 1 | 1 | 1 |
|---|---|---|
| 3 | 2 | 3 |
| A | B | C |

$3+1+3 = 7$